Introduction to Calculus B.Tech(HI) Winter Sem: 2022-2023

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Inverse substitution rule

- **Inverse Substitution Rule** If $\int f(g(t))g'(t)dt = F(t)+C$ and $x = g(t)$ is differentiable and invertible. Then

$$\int f(x)dx = \int f(g(t))g'(t)dt = F(t)+C = F(g^{-1}(x))+C.$$

Example: trigonometric substitutions

- **Ex.** Evaluate $\int \sqrt{a^2 - x^2}\,dx \quad (a > 0).$

- **Sol.** Let $x = a\sin t$, then $dx = a\cos t\,dt$.

$$\int \sqrt{a^2 - x^2}\,dx = \int a\cos t \cdot a\cos t\,dt = \int a^2\cos^2 t\,dt$$

$$= \frac{a^2}{2}\int (1+\cos 2t)\,dt = \frac{a^2}{2}\left(t + \frac{\sin 2t}{2}\right) + C$$

From $x = a\sin t$, we derive $t = \arcsin\dfrac{x}{a}$, $\quad \cos t = \dfrac{\sqrt{a^2 - x^2}}{a}$.

$$\int \sqrt{a^2 - x^2}\,dx = \frac{a^2}{2}\arcsin\frac{x}{a} + \frac{x\sqrt{a^2 - x^2}}{2} + C.$$

Example: trigonometric substitutions

- **Ex.** Evaluate $\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}} \quad (a>0).$

- **Sol.** Let $x = a\tan t$, then $dx = a\sec^2 t\,dt.$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \int \frac{a\sec^2 t}{a\sec t}dt = \int \sec t\,dt = \frac{1}{2}\ln\left|\frac{1+\sin t}{1-\sin t}\right| + C$$

From $x = a\tan t$, we derive $\sin t = \dfrac{x}{\sqrt{x^2+a^2}}$.

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \frac{1}{2}\ln\frac{(\sqrt{x^2+a^2}+x)^2}{a^2} + C = \ln\frac{\sqrt{x^2+a^2}+x}{a} + C.$$

Example

- **Ex.** Evaluate $\displaystyle\int \frac{\sqrt{9-x^2}}{x^2}\,dx.$

- **Sol.** $\displaystyle\int \frac{\sqrt{9-x^2}}{x^2}\,dx = \int \frac{\sqrt{9-(3\sin t)^2}}{(3\sin t)^2}\,d(3\sin t) = \int \frac{\cos^2 t}{\sin^2 t}\,dt$

$$= \int (\csc^2 t - 1)\,dt = -\cot t - t + C = -\frac{\sqrt{9-x^2}}{x} - \arcsin\frac{x}{3} + C.$$

Example

- **Ex.** Evaluate $\displaystyle\int \frac{1}{x^2\sqrt{x^2+4}}\,dx.$ substitution $x = 2\tan t$

- **Ex.** Find $\displaystyle\int \frac{1}{x^2\sqrt{x^2-4}}\,dx.$ substitution $x = 2\sec t$

- **Ex.** Find $\displaystyle\int \frac{1}{x\sqrt{3-2x-x^2}}\,dx.$

- $3-2x-x^2 = 4-(x+1)^2$ substitution $x = -1+2\sin t$

Example: reciprocal substitution

- **Ex.** Evaluate $\int \frac{dx}{x\sqrt{3x^2-2x-1}}$.
- **Sol.** Let $x=\frac{1}{t}$, then $dx=-\frac{1}{t^2}dt$.

$$\int \frac{dx}{x\sqrt{3x^2-2x-1}} = -\int \frac{dt}{\sqrt{3-2t-t^2}} = -\int \frac{dt}{\sqrt{2^2-(t+1)^2}}$$

$$= -\frac{1}{2}\int \frac{dt}{\sqrt{1-(\frac{t+1}{2})^2}} = -\arcsin\frac{t+1}{2}+C = -\arcsin\frac{x+1}{2x}+C.$$

Example: rational substitution

- **Ex.** Evaluate $\int \frac{x}{1+\sqrt{x}}dx.$

- **Sol.** Let $x=t^2,$ then $dx=2tdt.$

$$\int \frac{x}{1+\sqrt{x}}dx = \int \frac{t^2}{1+t}2tdt = 2\int \frac{t^3}{1+t}dt = 2\int \frac{t^3+1-1}{1+t}dt$$

$$= 2[\int (t^2-t+1)dt - \int \frac{1}{1+t}dt] = \frac{2t^3}{3} - t^2 + 2t - 2\ln(1+t) + C$$

$$= \frac{2\sqrt{x^3}}{3} - x + 2\sqrt{x} - 2\ln(1+\sqrt{x}) + C.$$

Inverse substitution for definite integral

- **The Inverse Substitution Rule** for definite integral: If *x*=g(t) is differentiable, invertible and, when *x* is in between *a* and *b*, t is in between $\alpha = g^{-1}(a)$ and $\beta = g^{-1}(b)$. Then

$$\int_a^b f(x)dx = \int_\alpha^\beta f(g(t))g'(t)dt.$$

Example

- **Ex.** Evaluate $\int_a^{2a} \frac{\sqrt{x^2-a^2}}{x^4}dx \quad (a>0).$

- **Sol.** Let $x = a\sec t,$ then $dx = a\sec t\tan t,$ and when x changes from a to 2a, t changes from 0 to $\pi/3.$

$$\int_a^{2a} \frac{\sqrt{x^2-a^2}}{x^4}dx = \int_0^{\frac{\pi}{3}} \frac{a\tan t}{a^4\sec^4 t} a\sec t\tan t dt$$

$$= \frac{1}{a^2}\int_0^{\frac{\pi}{3}} \sin^2 t\cos t dt = \frac{1}{3a^2}\sin^3 t\Big|_0^{\frac{\pi}{3}} = \frac{\sqrt{3}}{8a^2}.$$

Example

- **Ex.** Evaluate $\int_2^3 \frac{x+1}{\sqrt{(4x-x^2)^3}}\,dx.$
- **Sol.** Since $\sqrt{(4x-x^2)} = \sqrt{4-(x-2)^2}$, let $x = 2+2\sin t.$

Then $dx = 2\cos t,$ and when x changes from 2 to 3, t changes from 0 to $\pi/6.$

$$\int_2^3 \frac{x+1}{\sqrt{(4x-x^2)^3}}\,dx = \int_0^{\frac{\pi}{6}} \frac{3+2\sin t}{(2\cos t)^3}\,2\cos t\,dt = \int_0^{\frac{\pi}{6}} \frac{3+2\sin t}{4\cos^2 t}\,dt$$

$$= \left[\frac{3}{4}\tan t + \frac{1}{2\cos t}\right]_0^{\frac{\pi}{6}} = \left(\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{3}\right) - \frac{1}{2} = \frac{7\sqrt{3}-6}{12}.$$

Example: application of substitution

- **Ex.** Find $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx.$

- **Sol.** Let $x = \frac{\pi}{2} - t,$ then

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx = -\int_{\frac{\pi}{2}}^{0} \frac{\cos t}{\cos t + \sin t} dt = \int_0^{\frac{\pi}{2}} \frac{\cos t}{\sin t + \cos t} dt$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx = \frac{1}{2}\left(\int_0^{\frac{\pi}{2}} \frac{\cos t}{\sin t + \cos t} dt + \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx\right) = \frac{\pi}{4}.$$

Example: application of substitution

- **Ex.** Find the definite integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+e^x} dx.$

- **Sol.** $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+e^x} dx = \int_{-\frac{\pi}{2}}^{0} \frac{\sin^2 x}{1+e^x} dx + \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+e^x} dx.$

By substitution $x = -t,$

$$\int_{-\frac{\pi}{2}}^{0} \frac{\sin^2 x}{1+e^x} dx = -\int_{\frac{\pi}{2}}^{0} \frac{\sin^2 t}{1+e^{-t}} dt = \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+e^{-x}} dx = \int_{0}^{\frac{\pi}{2}} \frac{e^x \sin^2 x}{1+e^x} dx.$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+e^x} dx = \int_{0}^{\frac{\pi}{2}} \frac{e^x \sin^2 x}{1+e^x} dx + \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+e^x} dx = \int_{0}^{\frac{\pi}{2}} \sin^2 x dx = \frac{\pi}{4}.$$

Integration of rational functions

- **Any** rational function can be integrated by the following two steps: a). express it as a sum of simpler fractions by **partial fraction** technique; b. integrate each partial fraction using the integration techniques we have learned.
- For example, since

$$\frac{x+5}{x^2+x-2}=\frac{x+5}{(x-1)(x+2)}=\frac{2}{x-1}-\frac{1}{x+2}$$

we have

$$\int\frac{x+5}{x^2+x-2}dx=\int(\frac{2}{x-1}-\frac{1}{x+2})dx=2\ln|x-1|-\ln|x+2|+C.$$

Technique for partial fraction

- Take any rational function $f(x) = \dfrac{P(x)}{Q(x)}$, where P and Q are polynomials.
- If the degree of P is less than the degree of Q, we call f a **proper fraction**.
- If f is improper, that is, degree of P greater than or equal to degree of Q, then (by long division) we must have

$$f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)},$$

where S and R are also polynomials and degree of R less than degree of Q.

Technique for partial fraction

- For example, by long division, we can derive

$$\frac{x^3+x}{x-1}=x^2+x+2+\frac{2}{x-1}.$$

- For the above reason, we only need to consider the **proper** rational functions.
- The next step is to **factor** the denominator Q(x). It can be shown that any polynomial can be factored as a product of linear factors (in the form *ax*+*b*) and **irreducible** quadratic factors (in the form ax^2+bx+c with $b^2-4ac<0$). For example, $Q(x)=x^4$-16=(x-2)(x+2)(x^2+4).

$$x^4+x^2+1=(x^2+x+1)(x^2-x+1)$$

Technique for partial fraction

- The third step is to express the proper rational function R(x)/Q(x) as a sum of partial fractions of the form

$$\frac{A}{(ax+b)^i} \quad \text{or} \quad \frac{Ax+B}{(ax^2+bx+c)^j}.$$

- These two kind of rational functions can be integrated as

$$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C, \quad \int \frac{1}{(ax+b)^k}\,dx = \frac{1}{a(1-k)}(ax+b)^{-k+1} + C$$

$$\int \frac{Ax+B}{(ax^2+bx+c)^k}\,dx = C\int \frac{d(ax^2+bx+c)}{(ax^2+bx+c)^k} + D\int \frac{F}{[1+(x+E)^2]^k}\,dx.$$

Example

- **Ex.** Find $I_n = \int \frac{1}{(x^2+1)^n} dx.$
- **Sol.** $I_n = \int \frac{dx}{(x^2+1)^n} = \frac{x}{(x^2+1)^n} - \int x d \frac{1}{(x^2+1)^n}$

$$= \frac{x}{(x^2+1)^n} + 2n\int \frac{x^2 dx}{(x^2+1)^{n+1}} = \frac{x}{(x^2+1)^n} + 2n\int \frac{(x^2+1-1)dx}{(x^2+1)^{n+1}}$$

$$= \frac{x}{(x^2+1)^n} + 2nI_n - 2nI_{n+1}.$$

$$I_1 = \arctan x + C.$$

$$\Rightarrow I_{n+1} = \frac{2n-1}{2n} I_n + \frac{x}{2n(x^2+1)^n}.$$

$$I_2 = \frac{\arctan x}{2} + \frac{x}{2(x^2+1)} + C.$$

Technique for partial fraction

- From the above analysis, we see that how to split a rational function into partial fractions is the **key step** to integrate the rational function.
- When Q(x) contains factor $(x-a)^k$, the partial fractions contain $$\frac{A_1}{x-a}+\frac{A_2}{(x-a)^2}+\cdots+\frac{A_k}{(x-a)^k}.$$
- When Q(x) contains irreducible factor $(x^2+px+q)^k$, the partial fractions contain $$\frac{B_1x+D_1}{x^2+px+q}+\frac{B_2x+D_2}{(x^2+px+q)^2}+\cdots+\frac{B_kx+D_k}{(x^2+px+q)^k}.$$

Example

- **Ex.** Find $\int \frac{x^2+2x-1}{2x^3+3x^2-2x}\,dx.$

- **Sol.** Since $2x^3+3x^2-2x = x(x+2)(2x-1)$, the partial fraction has the form

$$\frac{x^2+2x-1}{2x^3+3x^2-2x} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{2x-1}.$$

Expanding the right side and comparing with the left side,

$$x^2+2x-1 = A(x+2)(2x-1) + Bx(2x-1) + Cx(x+2).$$

$$= (2A+2B+C)x^2 + (3A-B+2C)x - 2A$$

$$\Rightarrow A = \frac{1}{2}, B = \frac{1}{5}, C = -\frac{1}{10}$$

Example

- **Ex.** Find $I=\int\frac{x^5-x^3-2x^2+7x}{x^4-x^2-2x+2}dx.$
- **Sol.**

$$\frac{x^5-x^3-2x^2+7x}{x^4-x^2-2x+2}=x+\frac{5x}{x^4-x^2-2x+2}$$

$$x^4-x^2-2x+2=(x-1)^2(x^2+2x+2)$$

$$\frac{5x}{x^4-x^2-2x+2}=\frac{A_1}{x-1}+\frac{A_2}{(x-1)^2}+\frac{Bx+D}{x^2+2x+2}$$

$$\Rightarrow A_1=\frac{1}{5},A_2=1,B=-\frac{1}{5},D=-\frac{8}{5}$$

$$I=\frac{x^2}{2}+\frac{1}{5}\ln|x-1|-\frac{1}{x-1}-\frac{1}{10}\ln(x^2+2x+2)-\frac{7}{5}\arctan(x+1)+C.$$

Remark

- There are **two methods** to find the coefficients in the partial fractions. One method is comparing the corresponding coefficients of polynomials on both sides; the other is taking some special values of *x* in the identity.
- For instance, in the last example, we have

$$5x = A_1(x-1)(x^2+2x+2) + A_2(x^2+2x+2) + (Bx+D)(x^2-2x+1)$$

$$x = 1 \Rightarrow 5 = 5A_2 \Rightarrow A_2 = 1$$

$$x = -1+i \Rightarrow -5+5i = (B+3D)+(7B-4D)i \Rightarrow B = -\frac{1}{5}, D = -\frac{8}{5}$$

$$x = 0 \Rightarrow 0 = -2A_1 + 2A_2 + D \Rightarrow A_1 = \frac{1}{5}$$

Example

- **Ex.** Find $I = \int \frac{x^4 - x^3 - x^2 - 3x}{(x^2-1)(x^2+1)^2} dx.$
- **Sol.**

$$\frac{x^4 - x^3 - x^2 - 3x}{(x^2-1)(x^2+1)^2} = \frac{A_1}{x-1} + \frac{A_2}{x+1} + \frac{B_1 x + D_1}{x^2+1} + \frac{B_2 x + D_2}{(x^2+1)^2}$$

$$\Rightarrow A_1 = -\frac{1}{2}, A_2 = -\frac{1}{2}, B_1 = 1, D_1 = 1, B_2 = 1, D_2 = -1$$

$$I = -\frac{1}{2}\ln|x+1| - \frac{1}{2}\ln|x-1| + \frac{1}{2}\ln(x^2+1) + \arctan x + \int \frac{x-1}{(x^2+1)^2} dx$$

$$= \frac{1}{2}\ln\left|\frac{x^2+1}{x^2-1}\right| + \frac{1}{2}\arctan x - \frac{x+1}{2(x^2+1)} + C.$$

Homework 18

- Section 7.2: 18, 24, 42, 44
- Section 7.3: 5, 6, 24, 27